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GUIDED WORKSHOP · PART 1 OF 2

Machine Learning for Trading in the Age of AI Agents

From market idea to a validated feature-and-label dataset

Stefan Jansen · *Machine Learning for Trading*, 3rd Edition

Build with me, then inspect the result

1. Keep the workshop repository and this deck open side by side.
2. Run each named notebook section when the green checkpoint slide appears.
3. Stop at the expected output and compare it with the slide before continuing.
4. Ask when your output differs. Do not debug silently while the session moves on.

The notebooks are prepared. The work today is understanding and checking each decision, not typing the entire pipeline from an empty file.

One session, one break, one strategy

Segment	Mode	Time
Workflow, factors, and stack	Slides + discussion	35 min
Inspect the ETF data	Guided `01_data.ipynb`	15 min
Build with a coding agent	Presenter-led demo	20 min
Engineer features and labels	Guided `02_features_labels.ipynb`	30 min
Break	One short break	**10 min**
Model, backtest, research agents, next steps	Part 2	100 min

What you will leave with

- A complete LightGBM trading strategy built on 100 ETFs and daily data from 2006 to 2025.
- A point-in-time feature-and-label dataset with a manually verified 21-day forward return.
- Walk-forward predictions evaluated with both naive and HAC-corrected inference.
- A paired zero-cost and cost-aware backtest using identical signals and rebalance dates.
- Reusable notebooks, data, slide decks, and briefs for both agent demonstrations.

Six decisions produce one backtest

Data › Alpha factors › Labels + validation › ML model › Cost-aware backtest › Live handoff

Running these decisions out of order changes what the final number measures.

Agents assist; you approve the research

Stage	An agent can assist	You remain responsible for
Alpha factors	Propose hypotheses, summarize evidence, compare variants	Choosing a plausible mechanism
Labels and validation	Flag leakage risk and conflicting horizons	Approving the target and time split
Model	Build a pipeline from a falsifiable specification	Verifying inputs, outputs, and evaluation
Backtest	Compare cost and turnover scenarios	Certifying the result
Live handoff	Draft controls and monitoring checks	Deciding whether to deploy

A testable factor begins as a specification

Field	Example: 252-day momentum
Mechanism	Persistent sector flows may continue before reversing
Formula	252-trading-day percent price change within each ETF
Availability	Uses closes available through the feature date
Expected direction	Higher momentum predicts a higher 21-day forward return
Failure state	Sharp turning points can reverse before the feature responds

Research agents expand the candidate set

A research agent can	The decision still requires you to judge
Propose factor hypotheses from a market observation	Whether a mechanism exists beyond sample correlation
Summarize evidence for a candidate	Whether that evidence transfers to this universe and horizon
Compare variants on a common panel	Whether the comparison itself is fair

The useful output is a smaller set of testable specifications, not a longer list of trading ideas.

Eight features cover four market properties

Family	Features used today	Property represented
Momentum	21, 63, 126, and 252-day returns	Persistence across one to twelve months
Volatility	21 and 63-day realized volatility	Current risk regime
Mean reversion	RSI over 14 days	Position within the ETF's recent range
Liquidity	21-day dollar-volume percentile rank	Relative tradability on the same date

GUIDED CHECKPOINT 1 · `NOTEBOOKS/01_DATA.IPYNB` · 15 MIN

Verify the dataset before using it

1. Run through **Sanity checks before we build anything**.
2. Confirm the symbol count and observed date range.
3. Continue through **Point-in-time eligibility**.
4. Stop after the eligible-symbol count by year.

> Expected: 100 symbols, 2006-01-03 through 2025-12-31; XLC is the only ETF with less than ten years of observations.

The file contains 100 ETFs across 20 years

100	2006	2025
ETF symbols	first workshop date	last workshop date

- The data ships with the repository, so the workshop does not depend on a live data service.
- Coverage differs by symbol: XLC begins in 2018, while many ETFs cover the full workshop window.
- File presence alone does not make an ETF eligible for every historical decision.

Source: executed output, student-repo/notebooks/01_data.ipynb

Point-in-time eligibility changes the universe

- The 100-symbol file was selected with a backward-looking liquidity rule.
- The eligibility table records whether each symbol cleared that rule in each calendar year.
- Applying it before feature construction retains **84.2%** of all observed (symbol, day) rows.
- A 2006 model must not trade an ETF that became investable years later.

Source: executed output, student-repo/notebooks/02_features_labels.ipynb

Three libraries carry the trading pipeline

Library	Responsibility in this workshop
`ml4t-engineer`	RSI and point-in-time feature construction
`ml4t-diagnostic`	Walk-forward cross-validation and HAC-corrected IC inference
`ml4t-backtest`	Event-driven execution with commissions and spread

pandas handles the panel, LightGBM estimates the return model, and Polars supplies the backtest engine's columnar inputs. Everything installs from the attendee repository.

PRESENTER-LED DEMONSTRATION · 20 MIN

A coding agent builds against a falsifiable brief

Watch the first attempt, inspect two failure-prone steps, and compare the result with the prepared notebooks.

Precision determines what the agent can verify

The live brief requires the agent to:

1. Apply point-in-time eligibility before any feature is built.
2. Build the named eight features using information available by the feature date.
3. Stop before executing the 21-day forward-return label.
4. Use walk-forward validation with `label_horizon=21`.
5. Stop before reporting naive and HAC-corrected IC statistics.

The complete brief stays in `docs/coding_agent_brief.md` for reuse after the workshop.

Two checkpoints expose plausible wrong answers

Stop	Inspect	Failure it can reveal
Before labels run	Return direction, shift direction, and manual check	A target attached to the wrong date
Before evaluation runs	Splitter, purge horizon, IC series, and HAC horizon	Future information or overstated significance

An error-free run does not establish that either step is correct.

GUIDED CHECKPOINT 2A · `NOTEBOOKS/02_FEATURES_LABELS.IPYNB` · 12 MIN

Apply eligibility and build eight features

1. Run through **Point-in-time eligibility, applied**.
2. Confirm that 84.2% of observed rows remain eligible.
3. Run **Features** and inspect the eight-column summary.
4. Identify which features operate within a symbol and which operates across symbols.

> Expected: four momentum features, two volatility features, RSI(14), and cross-sectional dollar-volume rank.

Feature construction uses two axes

Within each ETF

- Momentum over four horizons
- Realized volatility over two horizons
- RSI with Wilder smoothing

Across ETFs on each date

- 21-day average dollar volume
- Percentile rank within the eligible cross-section

Mixing the two axes changes the meaning of the liquidity feature.

One line attaches the future outcome

```
g["fwd_ret_21d"] = g["close"].pct_change(21).shift(-21)
```

- `pct_change(21)` compares the current close with the close 21 rows earlier.
- `shift(-21)` attaches that realized return to the feature row 21 trading days earlier.
- The sign and shift direction require a price-level check; code review alone is insufficient.

GUIDED CHECKPOINT 2B · `NOTEBOOKS/02_FEATURES_LABELS.IPYNB` · 10 MIN

Prove the label against observed prices

1. Run **Labels: forward returns**.
2. Run **Prove the label doesn't leak**.
3. Compare the manual and computed SPY return.
4. Stop when the assertion passes.

> Expected: manual `0.002639`, computed `0.002639`, followed by the no-off-by-one confirmation.

Random splitting violates the trading clock

- Consecutive 21-day labels share roughly 20 days of their return paths.
- A random split scatters overlapping observations across training and validation sets.
- Walk-forward validation keeps training before testing.
- Purging removes training labels whose outcome windows cross the validation boundary.

The label horizon must agree with both the purge rule and the later HAC correction.

GUIDED CHECKPOINT 2C · `NOTEBOOKS/02_FEATURES_LABELS.IPYNB` · 8 MIN

Save the model-ready panel

1. Run **Assemble and save the model dataset**.
2. Confirm the final row count, symbol count, and date range.
3. Verify that `data/model_dataset.parquet` now exists.

> Expected: 369,159 rows, 99 symbols, 2008-01-03 through 2025-12-01.

Part 1 produced a falsifiable dataset

1. The universe is filtered at the decision date before features are built.
2. Each feature identifies its computation axis and available information.
3. The forward-return label matches a manual price calculation.
4. The saved panel is ready for leakage-aware walk-forward evaluation.

10-MINUTE BREAK

Save your notebook state

We resume with `notebooks/03_model_training.ipynb` and build out-of-sample predictions.